PCB Design Report

STM32 Ethernet + CAN + RS-485 Industrial Gateway

(Round 2, Problem 3)

Submitted By-

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EDA Tool: **EasyEDA**

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1. Introduction

This report describes a compact industrial gateway PCB that bridges Ethernet, CAN and RS-485. It has an STM32F407 MCU, a DP83848 Ethernet PHY with RJ45 magnetics, CAN, RS-485 and RS-232 transceivers, a USB-UART bridge, memory, a 24 V input with LM2596 buck stage and a 3.3 V LDO. Protection, termination and EMI-aware routing are applied throughout.

2. Component List (BOM)

Table 1 lists all components with designator, value, package and source. Parts from the EasyEDA library are marked "Library"; parts built from datasheet dimensions are marked "Custom".

Designator	Component	Value	Package	Source
D8	MCU STM32F407VET6	-	LQFP-100	Library
D10	Ethernet PHY DP83848IVV	-	LQFP-48	Library
D3	CAN transceiver SN65HVD230QDR	-	SOIC-8	Library
D4	USB-UART FT230XS-U	-	SSOP-16	Library
D6	RS-485 transceiver ST3485	-	SOIC-8	Library
D9	RS-232 transceiver ADM3202ARNZ	-	SOIC-16	Library
D1	SPI flash AT45DB161E	-	SOIC-8	Library
D2	EEPROM M95256	-	SOIC-8	Library
D5	Buck regulator LM2596S-5.0	5 V	TO-263-5	Library
D7	LDO LM1117IMPX-3.3	3.3 V	SOT-223	Library
VT1	NPN MMBT3904	-	SOT-23	Library
BQ1, BQ2, BQ3	32.768 kHz FC-135, 8 MHz HC49SM, CFPS-73	-	[packages]	[Library / Custom]
VD1, VD2, VD5, VD8	TVS/ESD SM15T39CA, SP0503BAHTG, SP0504BAHT	-	[SMB / SOT-143 / SOT-23-6]	[Library / Custom]
VD3, VD4, VD6, VD7	Diodes VS-30BQ040, MBR130LSFT1G, SS34, BAT54C	-	[SMC / SOD-123 / SMA / SOT-23]	Library
F1, F2	Resettable fuses MF-SM075/60, MF-MSMF050-2	-	[1812 / 1206]	Library
L1, L2	Inductor SRR1208-680YL, CM453232-101KL	68 uH / [value]	[SMD]	[Library / Custom]
G1	Buzzer HC0905A	-	[SMD]	[Library / Custom]
HL1-HL6	LEDs	-	0805	Library
R1-R70, C1-C54	Resistors and capacitors	[values]	0402 / [bulk caps]	Library
X1	USB Mini-B connector	-	SMD	Library
X2	Power input terminal (2-pin)	-	[pitch]	Library
X4	microSD socket	-	Push-push	Library
X5, X7	DB9 connector DRB-9FA	-	THT	[Library / Custom]
X12	RJ45 with magnetics TRJ0011TBNL	-	THT	[Library / Custom]
X3, X6, X8, X9, X10	Header pins (1x2, 2x3, 2x4, 1x4, 16-pin)	2.54 mm	THT	Library
S1, S2	Tactile switch / 8-position DIP switch	-	[SMD / THT]	Library
J1	Battery holder BK-912-TR	-	[package]	[Library / Custom]
X11	JST B16B-PHDSS 2x8 connector	-	[package]	[Library / Custom]

Table 1: Bill of materials

3. Schematic

The schematic is split into these parts: STM32 core with decoupling and crystals, Ethernet PHY with RJ45, CAN, RS-485 and RS-232 interfaces, USB-UART, SPI flash and EEPROM, and the 24 V input with buck and LDO supply. Net labels such as ETH_TX+, CAN_H and RS485_A connect all parts.

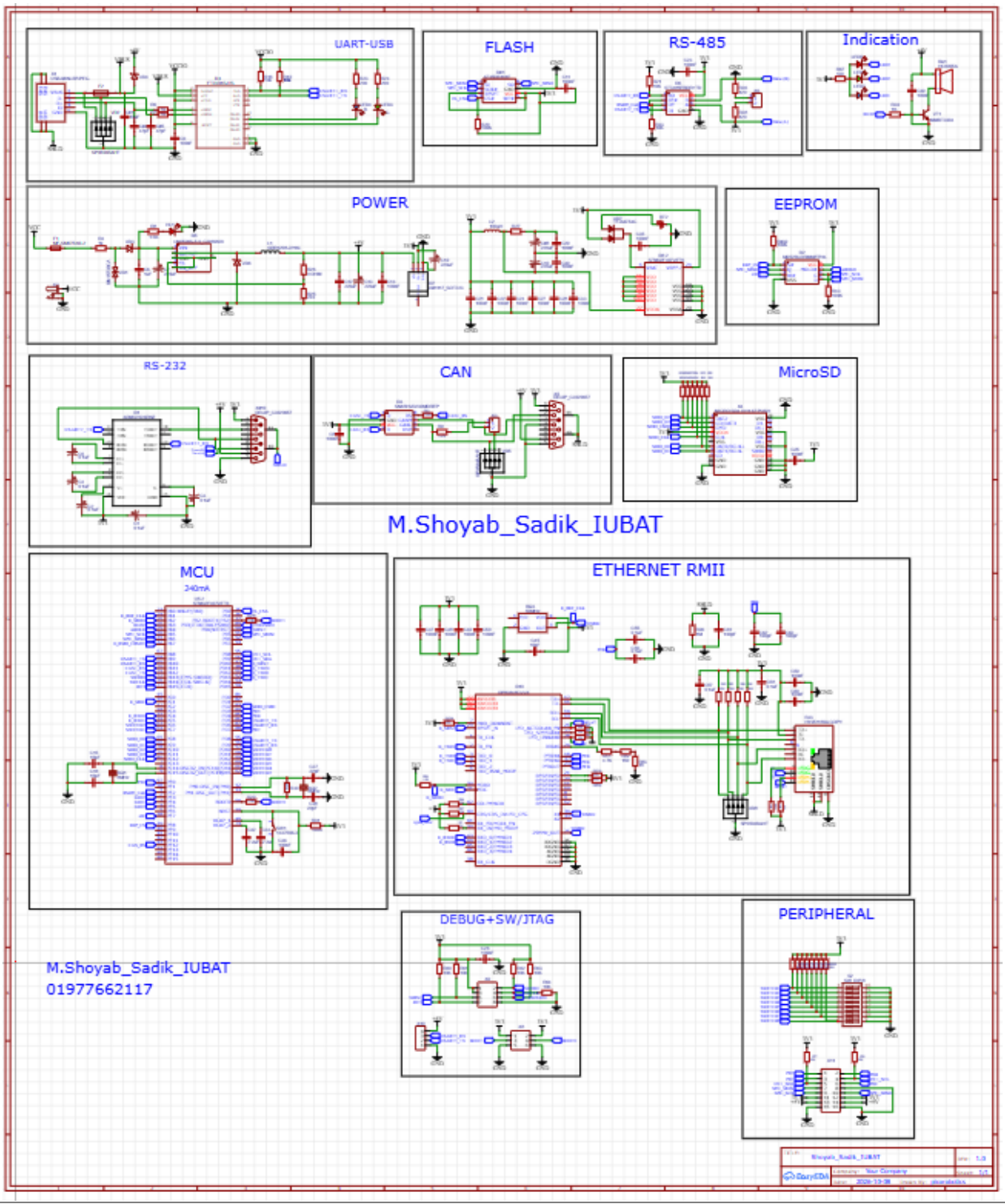


Fig1: Schematic of PCB

4. PCB Design Decisions

The board is [width: 120 x height: 100] mm with 2 layers. The PHY sits between the MCU and the RJ45. Connectors are on the board edges. The 24 V input and buck stage are in one corner, away from Ethernet and the crystals.

Requirement	How it was met
Ethernet TX/RX differential routing	[100 ohm pairs, length X mm, no vias, matched within X mm]
PHY-to-RJ45 placement	[D10 within X mm of X12, magnetics inside the RJ45]
CAN and RS-485 protection	[TVS next to the connector, series resistors, common-mode choke]
Termination resistor placement	[120 ohm resistors at bus connectors, jumper selectable]
Isolation / ground partitioning	[analog/digital/power GND regions joined at one point, split under the RJ45 chassis]
24 V input and buck stage	[fuse, TVS, D5 loop compact, wide traces, thermal copper at D5]
MCU decoupling	[100 nF at every VDD pin of D8, bulk caps near]
Connector placement	[X5, X7, X12, X1, X2 on edges]
EMI-aware routing	[solid ground return, crystal guard, short high-speed traces]
Test points and clean silkscreen	[test points on 24V, 5V, 3.3V, GND, signals; labels]

Table 2: Requirements checklist

5. Design Rules

Signal traces are 0.25 mm and clearance is 0.30 mm. Ethernet pairs are 100 ohm differential and the 24 V and buck paths are wider. A GND pour is used on both layers with stitching vias.

Rule	Value
Signal trace width	0.25 mm
Power trace width (24V, 5V, 3.3V)	0.25 mm
Ethernet differential pair	100 ohm
Clearance	0.30 mm
Via (hole / diameter)	0.35 mm
Copper pour	GND, top and bottom, stitching vias

Table 3: Design rules used

6. PCB Layout Views

Figure 2 shows the PCB design view and Figures 3, 4 and 5 show the 2D top layer, the 2D bottom layer and the 3D view respectively.

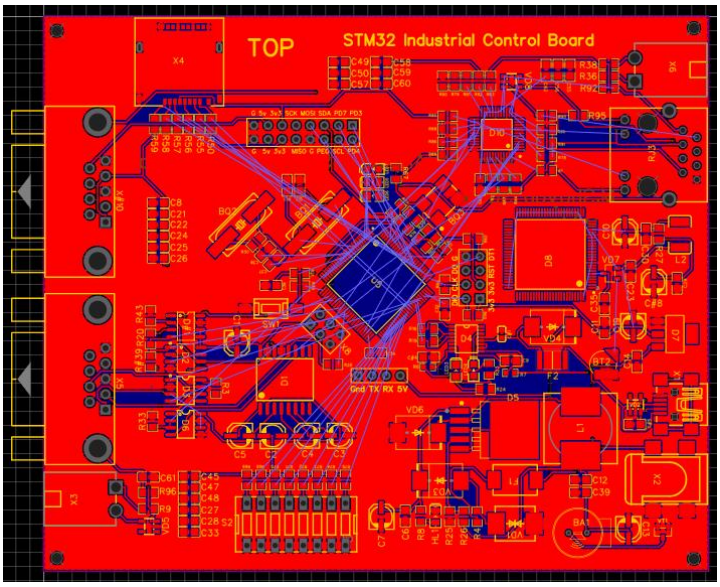


Fig2: PCB Design View

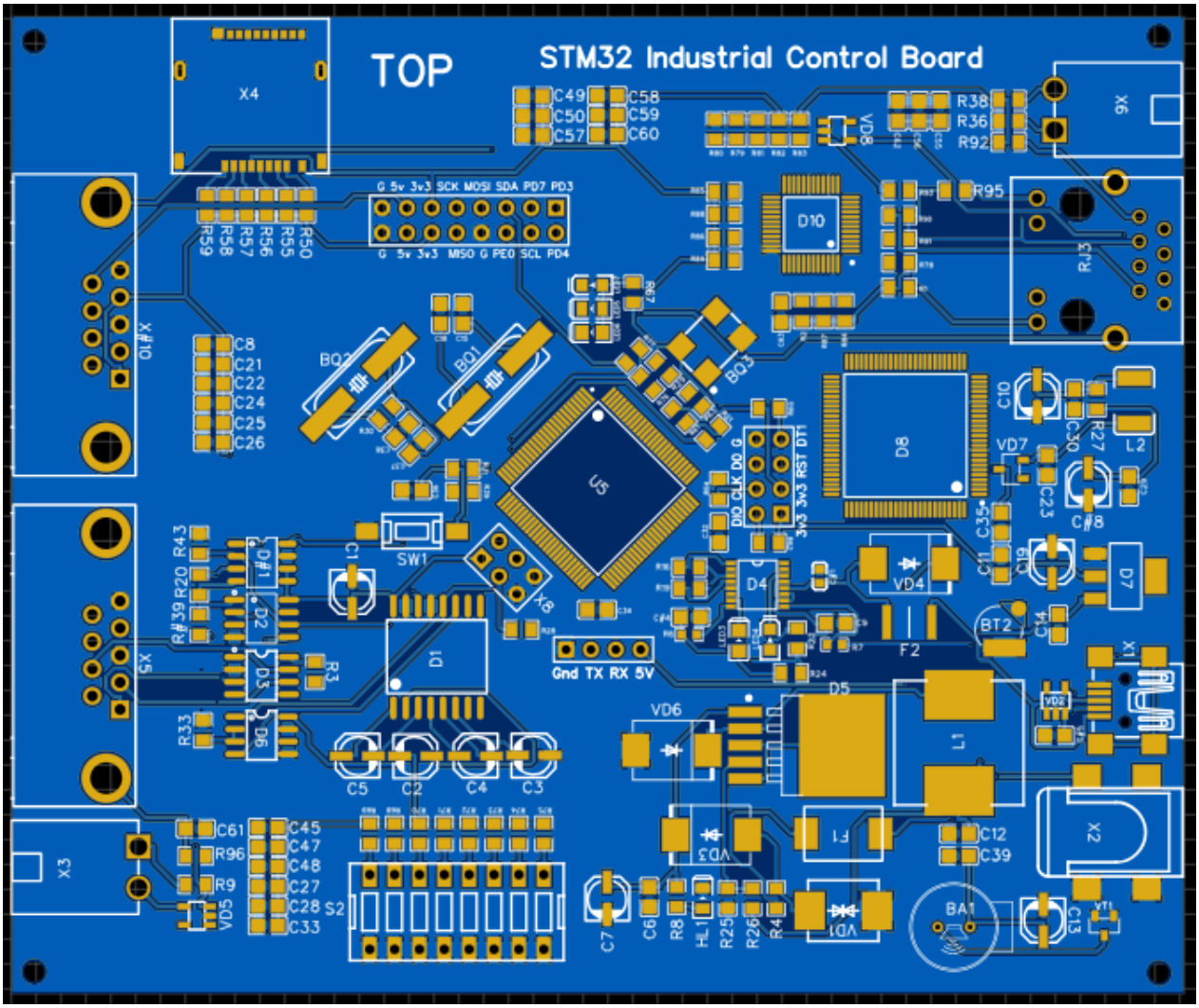


Fig3: Top View of PCB

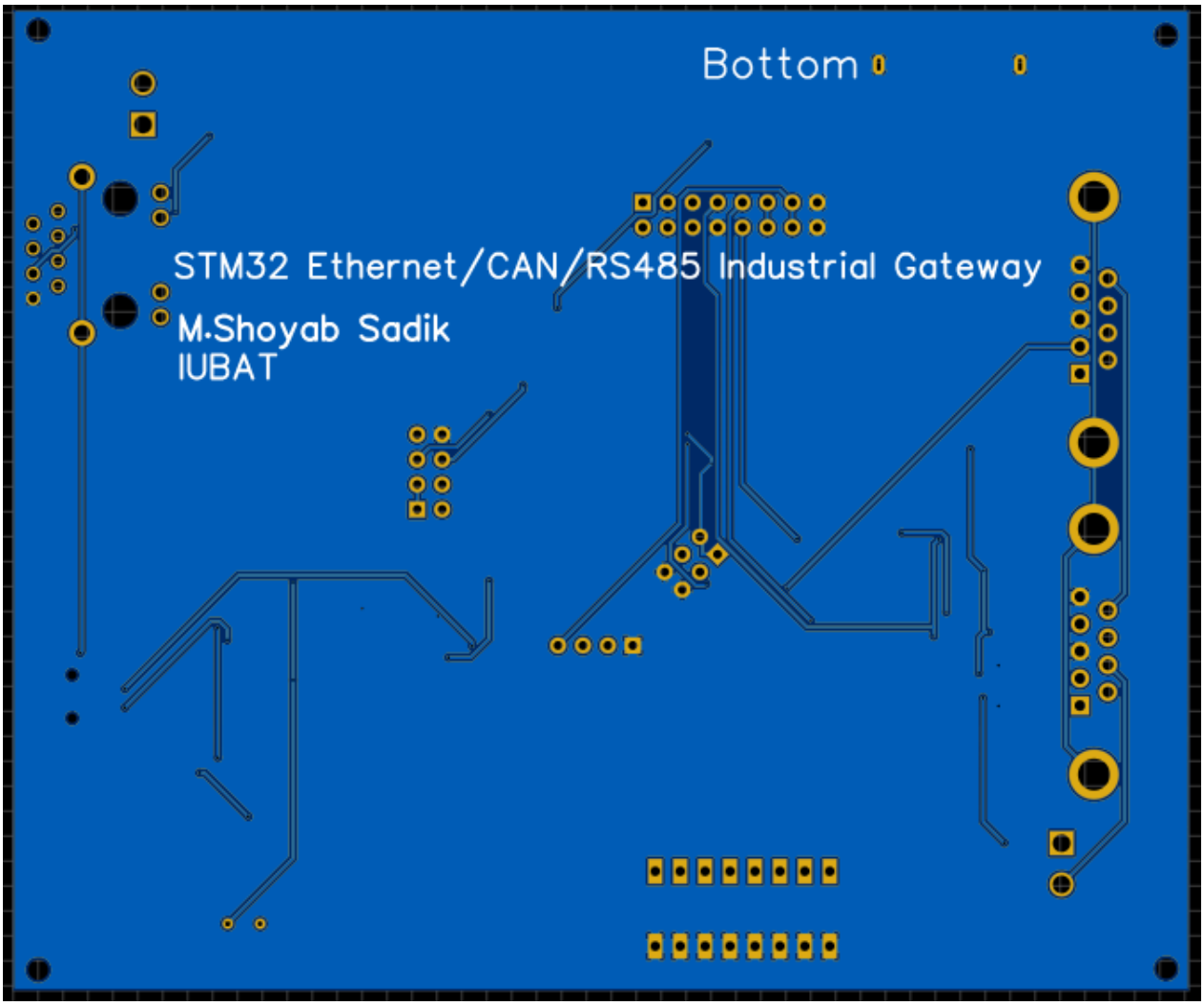


Fig4: Bottom View of PCB

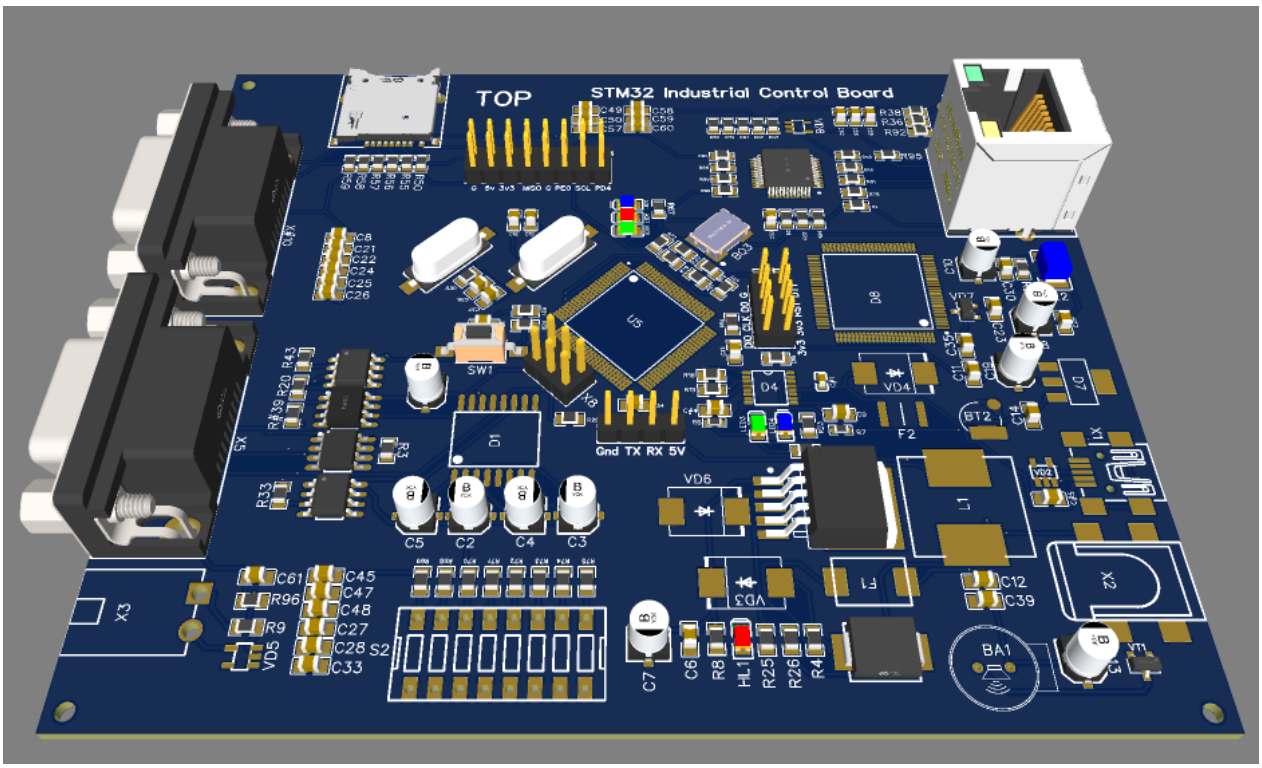


Fig5: 3D View of PCB

7. Conclusion

The design meets the stated requirements: clean Ethernet routing, PHY next to RJ45, protected CAN and RS-485, proper termination, ground partitioning, compact buck stage, decoupled MCU, clear connectors and silkscreen, and a clean DRC.

8. Files Submitted

- ShoyabSadik_IUBAT schematic file
- ShoyabSadik_IUBAT board file
- ShoyabSadik_IUBAT pdf (all layers, one per page)
- ShoyabSadik_IUBAT report (this document)

9. Files Source:

- GitHub Link: https://github.com/shoyabsadik/iTest-Bangladesh-PCB-Design-Contest-2026